

How to Train Your Intercept

A Comparison of Intercept Update Strategies in Coordinate Descent

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Abstract

Training intercepts in regularized regression is a common problem, yet it is often overlooked in the literature. This document explores different strategies for updating intercepts in coordinate descent algorithms, comparing gradient, Newton, and exact strategies.

Keywords: intercepts, coordinate descent, optimization, regularization

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4 1 Introduction

5 Intercept updating is a overlooked issue in the field of statistical learning. For regularized regression,
6 coordinate descent is the dominating algorithm, yet it is not clear how to best update the intercept
7 term. Implementations differ in how they handle this, for instance `skglm` uses a gradient step,
8 `LIBLINEAR` uses a Newton step, and `blitz` uses an exact update (based on Newton steps).

9 References

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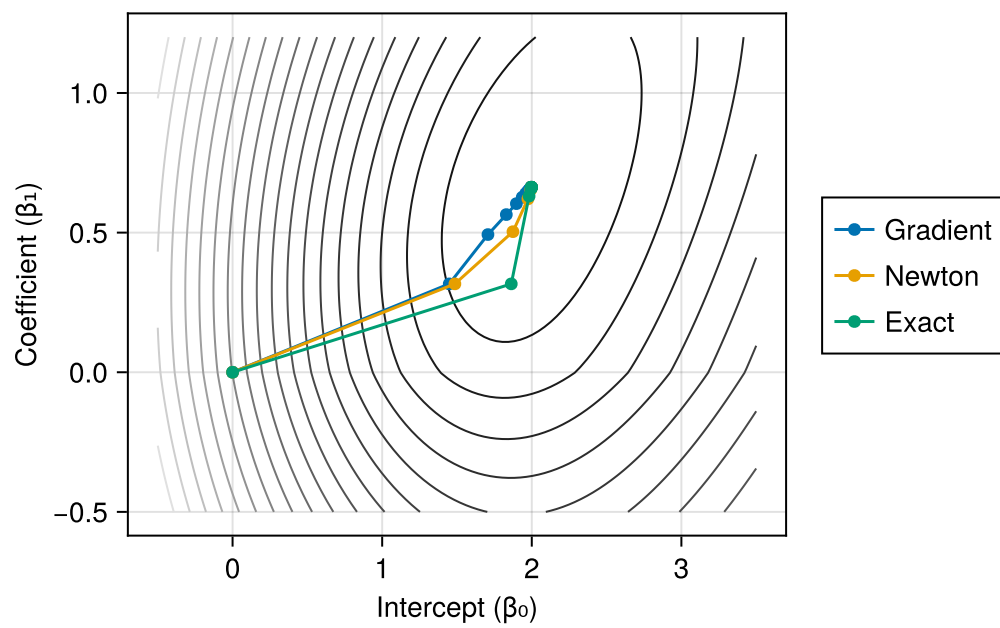


Figure 1: Parametric Plos